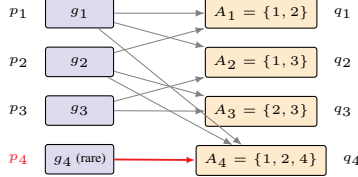


## SEC. 3 PROPOSED APPROACH



**Fig. 1.** The mask  $E$ . Batch compositions carry the mass  $q$  they occur with; groups must receive the mass  $p$  the objective assigns them. Group 4 is reachable only through  $A_4$ , so exact alignment requires  $p_4 \leq q_4$ : the Hall condition of Thm. 1 read on the singleton  $I = \{4\}$ . Shrinking the batch deletes  $A_4$  from the support and makes  $p_4$  unreachable.

### 1. PROPOSED APPROACH: ALIGNMENT BY MASKED OPTIMAL TRANSPORT

**Weights as a coupling.** Let  $Y[i, j] \geq 0$  be the weight assigned to group  $i$  when the active set is  $A_j$ , with  $Y[i, j] = 0$  for  $i \notin A_j$ , and let  $T[i, j] \triangleq q_j Y[i, j]$  be the joint mass carried from composition  $j$  to group  $i$ . Requiring that the weights be a probability vector on each realized composition, and that each group receive in expectation the mass  $p$  assigns it, is the *masked optimal transport* (MOT) feasibility problem of [6],

$$T\mathbf{1}_M = p, \quad \mathbf{1}_N^\top T = q, \quad T \geq 0, \quad \text{supp}(T) \subseteq E, \quad (1)$$

with mask  $E = \{(i, j) : i \in A_j\}$  encoding that a group absent from the batch cannot be weighted. Unlike classical transport [25, 20] there is no cost to minimize: any coupling respecting (1) serves, and (1) is the Kantorovich problem with the indicator cost  $C[i, j] = \infty$  off  $E$ . Fig. 1 is the picture.

**When alignment is possible.** Existence of a coupling is a supply–demand question on the bipartite graph  $E$ , and is settled by flow duality.

**Theorem 1** (Feasibility, [6, Thm. 1]). *For  $I \subseteq [N]$  let  $\mathcal{N}(I) = \{j : \exists i \in I, (i, j) \in E\}$  and  $\mathcal{S}(I) = \{j : A_j \subseteq I\}$ . Problem (1) is feasible if and only if*

$$\sum_{j \in \mathcal{S}(I)} q_j \leq \sum_{i \in I} p_i \leq \sum_{j \in \mathcal{N}(I)} q_j \quad \forall I \subseteq [N]. \quad (2)$$

Proofs are in the appendix. Read on singletons, (2) becomes a statement one can act on before training starts.

**Corollary 2** (Coverage law and severity). *Feasibility of (1) requires  $\mathbb{P}[A^t = \{i\}] \leq p_i \leq r_i$  for every  $i$ , with  $r_i = \mathbb{P}[i \in A^t]$  the observation rate of Sec. 2. Writing*

$$\nu \triangleq \sum_{i : p_i > r_i} p_i \quad (3)$$

#### Algorithm 1 GAVOT: group-fair SGD by masked transport

**Require:** target  $p$ , composition law  $(q, \{A_j\}, E)$ , step  $\eta$ , horizon  $T$ , CVaR parameters  $(\alpha, \gamma)$  with  $\gamma = 1$  for the risk-neutral method

- 1:  $(T, Y) \leftarrow$  Sinkhorn scaling for  $(p, q, E)$  ▷ offline, once
- 2:  $\theta^0 \in \mathcal{C}$ ,  $t^0 \in \mathbb{R}$
- 3: **for**  $t = 1, \dots, T$  **do**
- 4: draw batch  $B^t$ ; set  $A^t$ , let  $j(t)$  be s.t.  $A_{j(t)} = A^t$
- 5:  $g_i^t \leftarrow$  stochastic subgradient of  $f_i$  from  $B^t \cap$  group  $i$
- 6:  $w_i^t \leftarrow Y[i, j(t)](\gamma + \frac{1-\gamma}{\alpha} \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$
- 7:  $\theta^t \leftarrow \Pi_{\mathcal{C}}(\theta^{t-1} - \eta \sum_{i \in A^t} w_i^t g_i^t)$
- 8:  $t^t \leftarrow t^{t-1} - \eta(1 - \gamma)(1 - \frac{1}{\alpha} \sum_{i \in A^t} Y[i, j(t)] \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$
- 9: **end for**

**Ensure:**  $\bar{\theta}_T = \frac{1}{T} \sum_{t=1}^T \theta^t$

for the infeasible mass,  $\nu = 0$  iff the upper constraints hold. Under i.i.d. sampling of  $b$  examples the upper constraint reads  $b \geq \log(1 - p_i)/\log(1 - \pi_i)$ ; under  $K$ -of- $N$  uniform group sampling it reads  $p_i \leq K/N$ .

**Severity.** Importance may not exceed observation rate: a rare group is reachable only if the batch is large enough to contain it often enough. The infeasible mass  $\nu$  is the axis we sweep: at  $\nu = 0$  exact alignment is available, and as  $\nu$  grows more of  $p$  sits outside the set of row marginals the mask can reach. A *feasible* and an *infeasible* regime are two points on this continuum, and Sec. 5 sweeps it instead.

**Computing the plan.** Given feasibility, a coupling is obtained by iterative proportional fitting (Sinkhorn scaling) on the masked support [18, 19, 20]: starting from any  $T^{(0)} > 0$  on  $E$  with  $\mathbf{1}^\top T^{(0)} = q$ , alternate

$$T \leftarrow \text{diag}(p \oslash T\mathbf{1}) T, \quad T \leftarrow T \text{diag}(q \oslash T^\top \mathbf{1}) \quad (4)$$

until  $\|T\mathbf{1} - p\|_1 \leq \varepsilon$ . Each sweep costs  $O(|E|)$  with  $|E| \leq NM$  and  $M$  the number of *realized* compositions rather than  $2^N$ ; the plan is computed once, offline, and the training loop adds one lookup and one multiply per group per step. Under (2) the iteration converges to a solution of (1) [18, 22], being block-coordinate ascent on the dual of the entropic problem  $\min\{\sum_E T \log T : T\mathbf{1} = p, \mathbf{1}^\top T = q\}$ . Since  $\mathbf{1}_N^\top T = q$  and  $Y = T \text{diag}(q)^{-1}$ , every column  $Y[:, j]$  is a probability vector on  $A_j$ , the property each bound below uses. Algorithm 1 is the resulting method;  $\gamma = 1$  is the risk-neutral rule analyzed here and  $\gamma < 1$  the risk-averse layer of Sec. 4.

**Analysis.** Everything rests on one identity, [6, Lem. 1] moved from model averaging to gradient averaging. Let  $g_i(\theta)$  be the stochastic subgradient of  $f_i$  computed from  $B^t \cap$  group  $i$  and  $\hat{g}(\theta) = \sum_{i \in A^t} Y[i, j(t)] g_i(\theta)$  the transport-weighted batch subgradient of Algorithm 1. Transport removes bias; it is not a variance reduction device.

**Lemma 3** (Unbiasedness). *Let  $T$  satisfy (1). Then  $\mathbb{E}[\hat{g}(\theta) \mid \theta] \in \partial F_p(\theta)$  for every  $\theta \in \mathcal{C}$ .*

*Proof.* Conditionally on  $A^t = A_j$ ,  $\mathbb{E}[g_i(\theta)] \in \partial f_i(\theta)$  for  $i \in A_j$ . Averaging over  $A^t \sim q$ ,  $\mathbb{E}[\hat{g}] = \sum_j q_j \sum_{i \in A_j} Y[i, j] \partial f_i(\theta) = \sum_i (\sum_j T[i, j]) \partial f_i(\theta) = \sum_i p_i \partial f_i(\theta)$ , the row-marginal constraint being used in the last step.  $\square$

**Lemma 4** (Second moment, free of coverage).  $\mathbb{E}\|\hat{g}(\theta)\|^2 \leq G^2$  for every  $\theta \in \mathcal{C}$ , with no dependence on  $|A^t|$  or on  $N$ .

*Proof.*  $Y[\cdot, j]$  is a probability vector on  $A_j$ , so  $\|\sum_i Y[i, j] g_i\|^2 \leq \sum_i Y[i, j] \|g_i\|^2$  by convexity of  $\|\cdot\|^2$ ; take expectations and use  $\sum_j T[i, j] = p_i$ :  $\mathbb{E}\|\hat{g}\|^2 \leq \sum_i p_i \mathbb{E}\|g_i\|^2 \leq G^2$ .  $\square$

**Theorem 5** (Rate). *Under Assumption 1, if (1) is feasible then Algorithm 1 with  $\gamma = 1$  and  $\eta = D/(G\sqrt{T})$  satisfies*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) \leq \frac{DG}{\sqrt{T}}. \quad (5)$$

The constant in (5) is that of full-participation SGD and is independent of how many groups appear in a batch: a composition covering two groups yields the same bound as one covering all  $N$ , though with larger realized variance. The proof uses only the two lemmas and never refers to  $|A^t|$ , which is why the rate of [6] transfers verbatim from aggregation of models to aggregation of gradients.

**Outside the feasible region.** When (2) fails, (1) has no solution and the iteration (4) does not diverge: its iterates accumulate at the coupling whose row marginal  $\hat{p} \triangleq T\mathbf{1}$  is the KL-closest to  $p$  over the reachable polytope  $\mathcal{R}(q, E) = \{T\mathbf{1} : T \geq 0, \mathbf{1}^\top T = q, \text{supp}(T) \subseteq E\}$  [22, 21, 20]. The column marginal is met exactly, so Lemmas 3 and 4 hold with  $p$  replaced by  $\hat{p}$ : Algorithm 1 is then unbiased SGD on the surrogate  $F_{\hat{p}}$ , the closest objective the sampler can reach, and the loss on (1) is a bias of exactly the shape of Proposition 2. On singletons  $\hat{p}_i \leq r_i$ , so the groups with  $p_i > r_i$  are truncated at their observation rate and their missing mass,  $\nu$  up to renormalization, is redistributed.

**Theorem 6** (Infeasible regime). *Let  $\hat{p}$  be the row marginal of the Sinkhorn limit,  $\theta^\dagger \in \arg \min_{\mathcal{C}} F_{\hat{p}}$ , and write  $\Phi(w) \triangleq F_p(\arg \min_{\mathcal{C}} F_w)$  for the  $p$ -risk of the minimizer of the  $w$ -weighted risk. Then*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) = \underbrace{\Phi(\hat{p}) - \Phi(p)}_{\text{alignment floor}} + \underbrace{\varepsilon_T(\hat{p})}_{\text{optimization residual}}, \quad (6)$$

where  $\varepsilon_T(\hat{p}) \triangleq \mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^\dagger)$  satisfies  $\mathbb{E}F_{\hat{p}}(\bar{\theta}_T) - F_{\hat{p}}(\theta^\dagger) \leq DG/\sqrt{T}$  and  $\varepsilon_T(\hat{p}) \rightarrow 0$  whenever  $\bar{\theta}_T \rightarrow \theta^\dagger$ . Group-blind averaging obeys (6) with  $\tilde{p}$  in place of  $\hat{p}$ . Writing  $\Delta_{\text{floor}} \triangleq \Phi(\tilde{p}) - \Phi(\hat{p})$  for the alignment gain, transport improves on group-blind averaging exactly when  $\Delta_{\text{floor}} > \varepsilon_T(\hat{p}) - \varepsilon_T(\tilde{p})$ , and every floor is bounded by  $\Phi(w) - \Phi(p) \leq 2M\|p - w\|_1$ .

Theorem 6 replaces the usual bias *bound* [6] by an exact *decomposition*, both terms of which are available before training:  $\hat{p}$  and  $\tilde{p}$  from  $(p, q, E)$ , and  $\Phi$  exactly on instances with a closed-form or convex frontier. The gain is fixed by the instance and decides the sign of the comparison; the residual is paid by the estimator and, as Sec. 5 shows, is a stepsize artifact. The  $\ell_1$  bound is the weaker object and we report both, because it turns out too loose to discriminate.